



**RV College of
Engineering®**

Installing eSim on Ubuntu 25.04

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Submission Task 4**

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Introduction

Setting up the Virtual Machine on google cloud:

1. Create an instance in <https://console.cloud.google.com/compute/instances>
 2. Make sure to select Ubuntu 25.04 in the boot disk
 3. Setting up desktop environment and chrome remote desktop - connect via ssh and follow
<https://cloud.google.com/architecture/chrome-desktop-remote-on-compute-engine#gnome>
 4. Connect to the VM at <https://remotedesktop.google.com/access>
-

Installing patched version (this project):

Repository link: <https://github.com/KittyBorgX/eSim/tree/installers>

Disclaimer: All the code has been bundled into a zip file located in “Ubuntu-25.04” directory in the repository since there are changes to the ‘nghdl’ package as well which cannot be included in this repository. Instructions for building are noted below.

1. Clone the forked repository and checkout the installers branch:

```
git clone https://github.com/KittyBorgX/eSim.git && cd eSim
git checkout installers
```

2. Move the zip folder to a desired location (just an example given below):

```
mv Ubuntu-25.04/eSim-2.5.zip /home/krishna/
```

3. Unzip the folder and proceed with normal installation:

```
unzip eSim-2.5.zip
cd eSim-2.5
chmod +x install-eSim.sh
./install.sh --install
```

4. Installation should proceed normally and successfully install on Ubuntu 25.04

Challenge #1: Base Installation

Error:

```
krishna@esim-krishna-rvce:~/eSim-2.5$ ./install-eSim.sh --install
Detected Ubuntu Version:
Unsupported Ubuntu version: 25.04 ()
krishna@esim-krishna-rvce:~/eSim-2.5$
```

Explanation:

The install script checks for the installed ubuntu version so adding 25.04 as a supported version should correct it. Additionally, create install-eSim-scripts/install-eSim-25.04.sh by running:

```
cp install-eSim-scripts/install-eSim-24.04.sh install-eSim-scripts/install-eSim-25.04.sh
```

Implemented Fix:

Code changed (install-eSim.sh):

```
30 # function to choose and run the appropriate script
31 run_version_script() {
32     SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/install-eSim-scripts"
33
34     # Decide script based on full version
35     case $VERSION_ID in
36         "22.04")
37         if [[ "$FULL_VERSION" == "22.04.4" ]]; then
38             SCRIPT="$SCRIPT_DIR/install-eSim-22.04.sh"
39         else
40             SCRIPT="$SCRIPT_DIR/install-eSim-23.04.sh"
41         fi
42         ;;
43         "23.04")
44             SCRIPT="$SCRIPT_DIR/install-eSim-23.04.sh"
45             ;;
46         "24.04")
47             SCRIPT="$SCRIPT_DIR/install-eSim-24.04.sh"
48             ;;
49         "25.04")
50             SCRIPT="$SCRIPT_DIR/install-eSim-25.04.sh"
51             ;;
52         *)
53             echo "Unsupported Ubuntu version: $VERSION_ID ($FULL_VERSION)"
54             exit 1
55             ;;
56     esac
```

Change the following code to bump the Ubuntu version and bump KiCad version to 9.0 (latest stable)

```
krishna@sim-krishna-rvce:~/eSim-2.1$ diff --color=always install-eSim-scripts/install-eSim-24.04.sh install-eSim-scripts/install-eSim-25.04.sh
115,117c115,117
<     if [[ "$ubuntu_version" == "24.04" ]]; then
<         echo "Ubuntu 24.04 detected."
<         kicadppa="kicad/kicad-8.0-releases"
---
>     if [[ "$ubuntu_version" == "25.04" ]]; then
>         echo "Ubuntu 25.04 detected."
>         kicadppa="kicad/kicad-9.0-releases"
124c124
<         read -p "Do you want to remove it and install KiCad 8.0? (yes/no): " response
---
>         read -p "Do you want to remove it and install KiCad 9.0? (yes/no): " response
135c135
<         echo "KiCad 8.0 is already installed."
---
>         echo "KiCad 9.0 is already installed."
```

Further optimization is to change the way we check if the KiCad PPA exists. Current grep doesn't match the deb822 specification. We change it to match the new spec:

```
145c145
<     if ! grep -q "^deb .${kicadppa}" /etc/apt/sources.list /etc/apt/sources.list.d/* 2>/dev/null; then
---
>     if ! grep -qr "${kicadppa}" /etc/apt/sources.list.d/* 2>/dev/null; then
```

Challenge #2 (a): NGHDL version mismatch

Error:

```
creating: nghdl/Example/PWM/
inflating: nghdl/Example/PWM/pwmdecrement.vhdl
inflating: nghdl/Example/PWM/pwmincrement.vhdl
creating: nghdl/install-nghdl-scripts/
inflating: nghdl/install-nghdl-scripts/install-nghdl-22.04.sh
inflating: nghdl/install-nghdl-scripts/install-nghdl-23.04.sh
inflating: nghdl/install-nghdl-scripts/install-nghdl-24.04.sh
creating: nghdl/src/
inflating: nghdl/src/Appconfig.py
inflating: nghdl/src/createKicadLibrary.py
inflating: nghdl/src/model_generation.py
inflating: nghdl/src/ngspice_ghdl.py
creating: nghdl/src/ghdlserver/
inflating: nghdl/src/ghdlserver/compile.sh
inflating: nghdl/src/ghdlserver/ghdlserver.c
inflating: nghdl/src/ghdlserver/ghdlserver.h
inflating: nghdl/src/ghdlserver/uthash.h
inflating: nghdl/src/ghdlserver/Utility_Package.vhdl
inflating: nghdl/src/ghdlserver/Vhpi_Package.vhdl
detected Ubuntu Version:
Unsupported Ubuntu version: 25.04 ()
```

Explanation:

Included zip “nghdl” doesn’t support 25.04. For fixing further errors, we unzip “nghdl” , change it and finally zip the fixed version after all errors are resolved.

Implemented Fix:

To begin, comment line 66 in install-eSim-25.04.sh which prevents rewriting of the “nghdl” directory during further executions of the installation script.

```
61
62 function installNghdl
63 {
64
65     echo "Installing NGHDL....."
66     # unzip -o nghdl.zip
67     cd nghdl/
68     chmod +x install-nghdl.sh
69
70     # Do not trap on error of any command. Let NGHDL script handle its own errors.
71     trap "" ERR
72
73     ./install-nghdl.sh --install      # Install NGHDL
74
75     # Set trap again to error_exit function to exit on errors
76     trap error_exit ERR
77
78     ngspiceFlag=1
79     cd ../
80
81 }
82
```

Similar process as the initial installation script - we add support for Ubuntu 25.04 in "install-nghdl.sh"

Copy and create install-nghdl-25.04.sh:

```
cp nghdl/install-nghdl-scripts/install-nghdl-24.04.sh  
nghdl/install-nghdl-scripts/install-nghdl-25.04.sh
```

```
31 run_version_script() {  
32     SCRIPT_DIR="$(cd "$(dirname "${BASH_SOURCE[0]}")" && pwd)/install-nghdl-scripts"  
33  
34     # Decide script based on full version  
35     case $VERSION_ID in  
36         "22.04")  
37             if [[ "$FULL_VERSION" == "22.04.4" ]]; then  
38                 SCRIPT="$SCRIPT_DIR/install-nghdl-22.04.sh"  
39             else  
40                 SCRIPT="$SCRIPT_DIR/install-nghdl-23.04.sh"  
41             fi  
42             ;;  
43         "23.04")  
44             SCRIPT="$SCRIPT_DIR/install-nghdl-23.04.sh"  
45             ;;  
46         "24.04")  
47             SCRIPT="$SCRIPT_DIR/install-nghdl-24.04.sh"  
48             ;;  
49         "25.04")  
50             SCRIPT="$SCRIPT_DIR/install-nghdl-25.04.sh"  
51             ;;  
52         *)  
53             echo "Unsupported Ubuntu version: $VERSION_ID ($FULL_VERSION)"  
54             exit 1  
55             ;;  
56     esac
```

The script is run again prompting further errors which are discussed below.

Challenge #2 (b): Fixing NGHDL

Error:

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
Installing Zlib1g-dev.....
zlib1g-dev is already the newest version (1:1.3.dfsg+really1.3.1-1ubuntu1).
zlib1g-dev set to manually installed.
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1
Installing Gtk Canberra modules.....
Package libcanberra-gtk-module is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source

Error: Package 'libcanberra-gtk-module' has no installation candidate

Error! Kindly resolve above error(s) and try again.

Aborting Installation...

krishna@esim-krishna-rvce:~/eSim-2.5$
```

Explanation:

libcanberra-gtk-module is for GTK+ 2.0 applications while libcanberra-gtk3-module is for GTK+ 3.0 applications. These 2 are conflicting packages and hence libcanberra-gtk-module can be removed entirely.

Implemented Fix:

```
krishna@esim-krishna-rvce:~/eSim-2.5$ diff --color=always nghdl/install-nghdl-scripts/install-nghdl-24.04.
sh nghdl/install-nghdl-scripts/install-nghdl-25.04.sh
68c68
<     sudo apt install -y libcanberra-gtk-module libcanberra-gtk3-module
---
>     sudo apt install -y libcanberra-gtk3-module
312c312
```

Finally, uncomment the “unzip -o nghdl” line to revert the temporary fix.

Challenge #3: LLVM

Error:

```
Changing directory to ghdl-4.1.0 installation
Configuring ghdl-4.1.0 build as per requirements
gcc (Ubuntu 14.2.0-19ubuntu2) 14.2.0
Copyright (C) 2024 Free Software Foundation, Inc.
This is free software; see the source for copying conditions.  There is NO
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.

Use full IEEE library
Build machine is: x86_64-linux-gnu
Unhandled version llvm 20.1.2

Error! Kindly resolve above error(s) and try again.

Aborting Installation...
```

Explanation:

Error originates from installing ghdl-4.1.0 which has support only till LLVM 18. We can upgrade to ghdl-5.1.1 which adds support for LLVM 20. To do this, we replace the tar file in nghdl/ghdl-4.1.0 to nghdl/ghdl-5.1.1

Implemented Fix:

To bundle the ghdl-5.1.1.tar.gz instead of ghdl-4.1.0-tar.gz :

Running this in the nghdl directory:

```
wget https://github.com/ghdl/ghdl/archive/refs/tags/v5.1.1.tar.gz -O ghdl-5.1.1.tar.gz
```

Change the version of ghdl in install-nghdl-scripts/install-nghdl-25.04.sh

```
krishna@esim-krishna-rvce:~/eSim-2.5$ diff --color=always nghdl/install-nghdl-scripts/install-nghd
l-24.04.sh nghdl/install-nghdl-scripts/install-nghdl-25.04.sh
23c23
< ghdl="ghdl-4.1.0"
---
> ghdl="ghdl-5.1.1"
68c68
```

Lastly, zip the changed “nghdl” so the changed version is included and used.

Note that once this is complete, all the errors are fixed and eSim runs on Ubuntu 25.04.

Working screenshot on Ubuntu 25.04:

